



# Blender keyboard shortcuts and viewport navigation

X Y Z

## 3D Navigation via Gizmo



### Click or drag the gizmo

**Axis label:** jump to that orthographic view.

**Matching empty circle:** jump to the opposite view.

Hover near the gizmo to reveal a circle; drag inside it to orbit.



### Use the four tool icons

**Magnifying glass:** drag up/down to zoom.

**Hand:** drag to pan up, down, left, or right.

**Camera:** toggle camera view.

**Grid:** toggle orthographic / perspective.

## 3D Navigation via Number Pad

### NAVIGATING WITH THE KEYBOARD'S NUMBER PAD



Numberpad corner keys jump to fixed views or their opposite. The even numbers orbit the view in that cardinal direction.

**+** zooms in.

**-** zooms out.

### NAVIGATING WITH THE MIDDLE MOUSE BUTTON

**Orbit** Drag with the **middle mouse button** to rotate the camera around your scene.

**Pan** Hold **Shift** and drag with the middle mouse button to slide the view.

**Tip:** If you lose your object, select it in the Outliner, click back in the viewport for focus, and press **.** on the numberpad to bring it back into view.

## Simple workflow

REPEAT THIS LOOP

1. Select your target (objects in **Layout**; vertex/edge/face when **Modeling**).  
Use **Shift** to add to existing selection.
2. Select action/transform type (**G**rab/**R**otate/**S**cale)
3. **Optional:** Constrain **X**/**Y**/**Z** axis
4. Move mouse (no mouse button) for amount and direction (no mouse button), using **Shift** for precision or **Ctrl** for increments, or type a numeric value.
5. **Enter** to commit, or **ESC** to cancel & revert

## Transform selected

MAKING CHANGES

When an object, face, edge, or vertex is selected, use these commands for basic changes.

**G Grab / move**  
Press **G** to move. Add **X**, **Y**, or **Z** to stay on one axis.  
**G** + **X** → move along X  
**Shift** makes smaller movements.

**R Rotate**  
Press **R** to rotate. Add **X**, **Y**, or **Z** to rotate around that axis.  
**R** + **X** → rotate around X **R** + **R** → free rotate

**S Scale**  
Press **S** to make the object larger or smaller. Add **X**, **Y**, or **Z** to stretch on one axis.  
**S** + **X** → stretch along X

**Finish or undo:** **Enter** commits a move, rotation, or scale. **Esc** backs out and reverts it.

## Undo or adjust

CORRECTING MISTAKES

**Ctrl** + **Z** **Undo** — reverse the last action.  
**Shift** + **Ctrl** + **Z** **Redo** — restore the action.  
**F9** **Adjust last operation** — reopen the last operation's settings.

**Remember:** red = X, green = Y, blue = Z. The colored gizmo labels match these axes.

## Modeling mode

TOP-ROW NUMBERS

In **Edit Mode**, the top-row number keys choose what part of a mesh you select:

**1** Points / vertices  
**2** Edges  
**3** Faces

**Easy to mix up:** numberpad **1** is a front view; top-row **1** is vertex selection in Edit Mode.

## Viewport controls

PANELS + TOOLS

**N** **Sidebar** on/off — the right-side panel for numerical tweaks to your most recent change.  
**T** **Toolbar** on/off — the tool shelf on the left side of the viewport.  
**Shift** + **A** **Add an object** — opens the Add menu.

### Learn More

Blender Manual  
[docs.blender.org/manual/en/5.2/](https://docs.blender.org/manual/en/5.2/)  
Animated Cheatsheet  
[hollisbrown.github.io/blendershortcuts/](https://hollisbrown.github.io/blendershortcuts/)